SEO for Web Application

IF3110 – Web-based Application Development

Search Engine Optimization

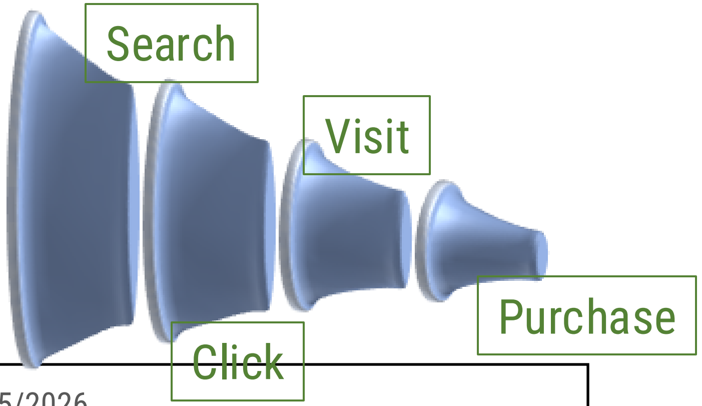
The process of improving your website's visibility in search engine results to attract more relevant traffic.

For Developers

- Technical implementation, not just marketing
- Code-level optimizations
- Performance considerations
- Structured data implementation
- Accessibility and crawlability

Why It Matters:

- 68% of online experiences begin with a search engine
- First page of Google captures 71% of traffic
- Technical SEO is YOUR responsibility



How Google Search Works

1. **CRAWLING**
   Googlebot discovers and fetches pages via links
2. **RENDERING**
   Executes JavaScript and processes content
   (Uses evergreen Chromium)
3. **INDEXING**
   Analyzes and stores content for search results

Crawling: How Googlebot Discovers Content

Discovery Methods

- Following links from other pages
- XML Sitemaps
- URL submissions via Search Console
- Social media and external sources

Crawling Process

1. Check robots.txt
2. Make HTTP request
3. Parse HTML for links
4. Add new URLs to crawl queue
5. Respect crawl rate limits

Rendering Page

- Traditional Site
  - HTML contains all contents immediately
- Modern Web Apps
  - Content generated by JavaScript after page load

Rendering Queues
- Pages wait for rendering resources
- Can take seconds to weeks
- Resource-intensive process

SSR – Googlebot sees (HTML) contents immediately

CSR – Empty HTML shell → JS execute → Content redered → Googelbot sees the content

Indexing

What Google Indexes

- Text content
- Images (with alt text)
- Videos
- Structured data
- Links and relationships
- Meta information

Ranking Factors

- Content relevance and quality
- Page experience (Core Web Vitals)
- Mobile-friendliness
- HTTPS security
- Structured data
- Backlinks

Fundamental Elements of SEO

- **robots.txt** - Control crawler access
- **Sitemap.xml** - Guide content discovery
- **URL Structure** - Clean, descriptive URLs
- **HTTP Status Codes** - Proper response codes
- **Meta Tags** - Page metadata
- **Canonicalization** - Handle duplicates

robots.txt Configuration

- Tell search engines which pages to crawl or avoid
- **Example robots.txt:**
  - User-agent: *
  - Disallow: /admin/
  - Disallow: /api/
  - Disallow: /temp/
  - Allow: /api/public/

  - Sitemap: https://example.com/sitemap.xml
- **Common Mistakes:**
  - Blocking CSS/JavaScript files
  - Blocking entire site accidentally
  - Using robots.txt for sensitive data

XML Sitemaps

- XML file listing all important URLs on your site
- **Example Structure:**

```xml
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <url>
    <loc>https://example.com/products</loc>
    <lastmod>2025-12-07</lastmod>
    <changefreq>weekly</changefreq>
    <priority>0.8</priority>
  </url>
</urlset>
```

- **Best Practices:**
  - Update regularly
  - Include only canonical URLs
  - Submit to Search Console
  - Split large sites (50,000 URL limit)

SEO-Friendly URLs

- **Good URL Structure:**
  - https://example.com/products/laptops/gaming
  - https://example.com/blog/seo-guide
  - https://example.com/about/team
- **Poor URL Structure:**
  - https://example.com/page?id=12345&cat=67
  - https://example.com/index.php?p=product&item=abc
  - https://example.com/2/6772756D707920636174
- **Best Practices:**
  - Use descriptive words
  - Separate with hyphens (-)
  - Keep it short and readable
  - Use lowercase
  - Avoid parameters when possible

URL Rewriting & SEO

- The process of converting dynamic, parameter-heavy URLs into clean, human-readable, and search engine-friendly URLs.
- **Example:**
  - Original: example.com/products.php?cat=1&id=123
  - Rewritten: example.com/products/electronics/smartphone-x
- **Best Practices:**
  - **Descriptive & Concise:** Include keywords, keep it short.
  - **Hyphens for Separators:** word-word
  - **Lowercase:** For consistency.
  - **301 Redirects:** Crucial for old URLs to new ones.
  - **Canonical Tags:** Use even with rewritten URLs to confirm preferred version.
- **Why it Matters for SEO:**
  - **Readability & User Experience:**
    - Easier to understand, remember, and share.
    - Increases trust and click-through rates.
  - **Keyword Inclusion:**
    - Allows relevant keywords directly in the URL path.
    - Provides a minor relevance signal to search engines.
  - **Crawlability & Indexing:**
    - Simpler URLs are easier for crawlers to process.
    - Helps prevent duplicate content issues when combined with canonicalization.
  - **Site Structure:**
    - Reflects logical hierarchy (e.g., /category/subcategory/product).

Communicating Page Status

- **Common Status Codes:**
  - **200 OK** - Page exists and is accessible
  - **301 Moved Permanently** - Permanent redirect
  - **302 Found** - Temporary redirect
  - **404 Not Found** - Page doesn't exist
  - **410 Gone** - Permanently removed
  - **503 Service Unavailable** - Temporary downtime
- **SEO Impact:**
  - 301: Passes ~90-99% of link equity
  - 404: Normal, not a penalty
  - Soft 404: Problematic (200 for missing content)

HTML Metadata for SEO

```html
<!-- Title Tag (Most Important!) -->
<title>SEO Guide for Web Developers | Example.com</title>

<!-- Meta Description -->
<meta name="description" content="Learn technical SEO
implementation for modern web applications. Complete
guide with code examples.">

<!-- Viewport (Mobile) -->
<meta name="viewport" content="width=device-width,
initial-scale=1.0">

<!-- Robots -->
<meta name="robots" content="index, follow">
```

Tag Best Practices

- **Formula:**
  - Primary Keyword | Secondary Keyword | Brand
- **Examples:**
  - Gaming Laptops - High Performance | TechStore
  - Learn React Hooks - Complete Tutorial | DevAcademy
  - SEO for Web Apps | IF3110 Course Materials
- **Guidelines:**
  - 50-60 characters (optimal)
  - Unique for each page
  - Include target keywords
  - Front-load important words
  - Avoid keyword stuffing

Handling Duplicate Content

- **The Problem:**
  - https://example.com/products
  - https://www.example.com/products
  - https://example.com/products?ref=home
  - https://example.com/products/
  - All show the same content!
- **The Solution:**
  - html
  - Copy
  - <link rel="canonical"
  - href="https://example.com/products">
- **When to Use:**
  - URL parameters (sorting, filtering)
  - HTTP vs HTTPS
  - www vs non-www
  - Trailing slashes
  - Duplicate content across pages

JavaScript SEO Challenges

- **The Challenge:**
  - Traditional crawlers see empty HTML shells
- **Example - What Googlebot Initially Sees:**

```html
<!DOCTYPE html>
<html>
<head><title>My App</title></head>
<body>
  <div id="root"></div>
  <script src="bundle.js"></script>
</body>
</html>
```

- **After JavaScript Execution:**
  - Full content appears, but requires rendering!

Single Page Applications (SPAs)

- **SPA Characteristics:**
  - One HTML page
  - Client-side routing
  - Dynamic content loading
  - Heavy JavaScript dependency
- **SEO Challenges:**
  - Content not in initial HTML
  - Routing with hash fragments
  - Soft 404 errors
  - Slow rendering
  - Broken back button
- **Solutions:**
  - Server-Side Rendering (SSR)
  - Static Site Generation (SSG)
  - Dynamic Rendering
  - Proper History API usage

Hash Fragments vs History API

Client-Side Routing

BAD: Hash Fragments

```
// URLs like: example.com/#/products
<a href="#/products">Products</a>
<a href="#/about">About</a>

// Googlebot cannot reliably crawl these!
```

✅ GOOD: History API

```
// URLs like: example.com/products
<a href="/products">Products</a>

// JavaScript
function navigate(url) {
  fetch(url).then(data => updateContent(data));
  window.history.pushState({}, '', url);
}
```

Handling Soft 404 Errors

The Problem – SPA returns 200 OK for non-existent pages

Solution 1: Redirect to 404 Page

```
fetch(`/api/products/${productId}`)
  .then(response => response.json())
  .then(product => {
    if(product.exists) {
      showProductDetails(product);
    } else {
      // Redirect to server 404 page
      window.location.href = '/not-found';
    }
  });
```

Solution 2: Add noindex Meta Tag

```
const metaRobots = document.createElement('meta');
metaRobots.name = 'robots';
metaRobots.content = 'noindex';
document.head.appendChild(metaRobots);
```

Server-Side Rendering (SSR)

- **How SSR Works:**
  - User requests page
  - Server executes JavaScript
  - Server generates full HTML
  - Browser receives complete content
  - JavaScript hydrates for interactivity
- **Benefits:**
  - Googlebot sees content immediately
  - Faster initial page load
  - Better social media sharing
  - Improved accessibility

Dynamic Meta Tags with JavaScript

- Updating Page Metadata
- **Setting Title and Description:**

```
// Update title
document.title = 'Product Name | My Store';

// Update or create meta description
let metaDesc = document.querySelector('meta[name="description"]');
if (!metaDesc) {
  metaDesc = document.createElement('meta');
  metaDesc.name = 'description';
  document.head.appendChild(metaDesc);
}
metaDesc.content = 'Product description here...';
```

- **Setting Canonical URL:**

```
const canonical = document.createElement('link');
canonical.rel = 'canonical';
canonical.href = 'https://example.com/products/laptop';
document.head.appendChild(canonical);
```

Lazy Loading & SEO

- **Image Lazy Loading:**

```html
<!-- Native lazy loading -->
<img src="product.jpg"
     alt="Gaming Laptop"
     loading="lazy">
```

- **Best Practices:**
  - Use native loading="lazy" for images
  - Don't lazy load above-the-fold content
  - Provide placeholder dimensions

- **JavaScript Lazy Loading:**

```javascript
// Intersection Observer API
const observer = new IntersectionObserver((entries) => {
  entries.forEach(entry => {
    if (entry.isIntersecting) {
      loadContent(entry.target);
    }
  });
});

document.querySelectorAll('.lazy').forEach(el => {
  observer.observe(el);
});
```

Helping Search Engines Understand Content

- **What is Structured Data -** Standardized format to provide information about a page and classify content
- **Benefits:**
  - Rich results in search
  - Featured snippets
  - Knowledge panels
  - Better click-through rates
  - Enhanced search appearance
- **Format: JSON-LD** (Recommended by Google)

```json
{
  "@context": "https://schema.org",
  "@type": "Article",
  "headline": "SEO Guide",
  "author": "John Doe"
}
```

Core Web Vitals

- Page Experience Metrics
- **Three Key Metrics:**
  - **LCP - Largest Contentful Paint -** Time until the largest content element is visible
    Loading performance
    Target: < 2.5 seconds
  - **FID/INP - First Input Delay / Interaction to Next Paint**
    - Time from first user interaction to browser response
    - Responsiveness throughout page lifetime (replacing FID)
      Interactivity
      Target: < 100ms (FID) / < 200ms (INP)
  - **CLS - Cumulative Layout Shift -** Measures unexpected layout shifts during page load
    Visual stability
    Target: < 0.1

Mobile-First Indexing

- **What Changed:**
  - Google primarily uses mobile version for indexing
  - Desktop version is secondary
  - Started rolling out in 2018
- **Requirements:**
  - Responsive design
  - Same content on mobile and desktop
  - Structured data on both versions
  - Meta tags on both versions
  - Fast mobile loading
  - Mobile-friendly navigation
- **Testing:**
  - Mobile-Friendly Test tool
  - Search Console mobile usability report
  - Chrome DevTools device emulation

Image Optimization for SEO

- **Technical Optimization:**

```html
<img src="laptop.webp"
     alt="Gaming laptop with RGB keyboard"
     width="800"
     height="600"
     loading="lazy">
```

- **Key Elements:**
  - **Alt Text** - Descriptive, keyword-rich
  - **File Format** - WebP > JPEG > PNG
  - **Dimensions** - Specify width/height
  - **Compression** - Balance quality/size
  - **Responsive Images** - Use srcset
  - **Lazy Loading** - Below-the-fold images

- **Responsive Images:**

```html
<img srcset="small.jpg 480w,
             medium.jpg 800w,
             large.jpg 1200w"
     sizes="(max-width: 600px) 480px,
            (max-width: 1000px) 800px,
            1200px"
     src="large.jpg" alt="Product">
```

Leveraging Google Search Console

- **Performance Report**
  - Impressions, clicks, CTR
  - Query analysis
  - Page performance
- **Coverage Report**
  - Indexed pages
  - Crawl errors
  - Excluded pages
- **URL Inspection Tool**
  - Check indexing status
  - View rendered page
  - Request indexing
- **Core Web Vitals**
  - Performance metrics
  - Mobile/desktop breakdown
- **Sitemaps**
  - Submit sitemaps
  - Track submission status

Pre-Launch SEO Checklist

- **Technical Foundation:**
  - HTTPS enabled (SSL certificate)
  - robots.txt configured correctly
  - XML sitemap created and accessible
  - Google Search Console set up
  - Google Analytics installed
- **On-Page SEO:**
  - Unique title tags (all pages)
  - Meta descriptions (all pages)
  - Heading hierarchy (H1, H2, H3)
  - Alt text for all images
  - Internal linking structure
- **Technical SEO:**
  - Mobile responsive design
  - Fast loading speed (< 3 seconds)
  - No broken links (404 errors)
  - Canonical URLs set
  - Structured data implemented

Common SEO Mistakes

- ❌ **Blocking Resources in robots.txt**
  - # DON'T DO THIS:
  - Disallow: /css/
  - Disallow: /js/
  - Google needs CSS/JS to render properly!
- ❌ **Using Hash Fragments for Navigation**
  - <a href="#/products"> <!-- Bad -->
  - <a href="/products"> <!-- Good -->
- ❌ **Duplicate Content**
  - Same content on multiple URLs
  - No canonical tags
- ❌ **Missing Alt Text**
  - <img src="product.jpg"> <!-- Bad -->
  - <img src="product.jpg" alt="Gaming laptop"> <!-- Good -->
- ❌ **Slow Page Speed**
  - Unoptimized images
  - No caching
  - Render-blocking resources

Reference

- **Continue Your SEO Journey**
- **Official Documentation:**
  📚 Google Search Central
  https://developers.google.com/search/docs
- 📚 SEO Starter Guide
  https://developers.google.com/search/docs/fundamentals/seo-starter-guide
- 📚 JavaScript SEO Guide
  https://developers.google.com/search/docs/crawling-indexing/javascript
- 📚 Structured Data Documentation
  https://developers.google.com/search/docs/appearance/structured-data
- **Tools:**
  🔧 Google Search Console
  🔧 PageSpeed Insights
  🔧 Rich Results Test
  🔧 Lighthouse (Chrome DevTools)